

Nguyen Dat Tho

EMBEDDED LINUX DEVELOPER

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”Building reliable embedded Linux systems from development to deployment.”

Summary

Embedded Linux Developer with 9+ years of experience designing and maintaining industrial-grade Linux distributions and platform software. Strong hands-on expertise in Yocto Project, Linux kernel and boot stack troubleshooting, CI/CD automation, and hardware-in-the-loop validation.

Proven track record delivering production systems for long-lifecycle products, including OTA update integration, reproducible builds, and Debian stable transitions. Focused on building reliable, maintainable, and secure embedded software with efficient development workflows.

Work Experience

Toshiba Software Development Vietnam (TSDV)	Hanoi, Vietnam
SOFTWARE ENGINEER PROJECT: DEVELOPMENT OF NV SERIES FOR INDUSTRIAL COMPUTER	Jan. 2015 - Mar. 2018
- Architected the migration of legacy C applications from 32-bit to 64-bit architecture, ensuring seamless compatibility and performance gains. - Customized and optimized Debian-based distributions using Yocto, satisfying strict industrial reliability requirements. - Engineered an automated testing framework using Fuego to ensure rapid and reliable delivery of OS updates. - Using QEMU-KVM virtualization with hardware binding to maximize performance for specialized industrial workloads.	
Toshiba Software Development Vietnam (TSDV)	Hanoi, Vietnam
SENIOR SOFTWARE ENGINEER PROJECT: DEVELOPMENT OF TOSHIBA SKERLET OS	Apr. 2018 - Mar. 2022
- Developed and maintained production-grade Linux OS using Yocto (Morty) and meta-debian. - Implemented automated hardware testing by deploying LAVA and Fuego, significantly reducing manual validation effort. - Resolved complex bootloader and kernel issues on diverse embedded platforms, including BeagleBone Black and Raspberry Pi. - Investigated and fixed several CVEs (Common Vulnerabilities and Exposures) to enhance system security.	
Toshiba Software Development Vietnam (TSDV)	Hanoi, Vietnam
SENIOR SOFTWARE ENGINEER PROJECT: DEVELOPMENT OF TOSHIBA LINUX OS (NEW GENERATION)	Apr. 2022 - Aug. 2024
- Developed and maintained modular Linux OS distributions using the Debos build tool. - Deployed Fossology to automate the management of open-source licenses and ensure compliance across the organization. - Architected containerized CI/CD pipelines using GitLab CI and Docker for cross-project efficiency. - Contributed to the CIP (Civil Infrastructure Platform) project, collaborating with Renesas and Siemens on long-term OS maintenance. - Innovated product features by integrating SWUpdate for secure, fail-safe OTA updates and implementing reproducible builds. - Managed the full-cycle transition of Linux OS distributions to newer Debian stable releases. (e.g., Buster to Bullseye, Bullseye to Bookworm) - Explored and benchmarked WebAssembly (WASM) runtimes across Ubuntu, Debian, Zephyr, and ROS.	
LG Electronics Development Vietnam (LGEDV)	Hanoi, Vietnam
SENIOR SOFTWARE ENGINEER PROJECT: WEBOS & ATLAS	Sep. 2024 - Present
- Managing the end-to-end SCM lifecycle for LG’s next-generation platform, leveraging Jenkins, Gerrit, and Grafana. - Reviewing and approving Yocto layers and recipes changes for integration new features into the mainline build. - Developing and optimizing Yocto-based build environments for high-performance Linux distributions. (ccache, sstate-cache...) - Bring-up and integration of new hardware platforms into the Yocto build system (QCS6490, NVidia Orin, DQ1) - Actively contributing technical improvements to the Yocto Project and OpenEmbedded upstream repositories (openembedded-core, meta-openembedded) - Implementing CVE & license system to monitor and manage security vulnerabilities and compliance. (Fosslight and CVE Tracker)	

Skills

Core Embedded	Linux Kernel Development, Yocto Project, Debos, meta-debian, KAS, SWUpdate, OTA Updates
Languages	C C++, Python, Bash, Go, YAML, CMake
Build & DevOps	Jenkins, Gerrit, GitLab CI, Docker, Grafana, Git, Fossology
Platforms	BeagleBone Black, Raspberry Pi, QCS6490, NVidia Orin, DQ1, Debian, Ubuntu, Zephyr, ROS
Testing	Fuego, LAVA, Automated Hardware-in-the-Loop (HIL) Testing